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SEMESTER S1

MATHEMATICS FOR INFORMATION SCIENCE – 1
(Group A)

Course Code	**GAMAT101**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in single variable calculus	**Course Type**	Theory

Course Objectives:

1. To provide students with essential skills in analyzing functions of several variables, identifying extrema, and optimizing processes to preparing them to address diverse engineering applications and challenges.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Limits of Function Values, Continuity at a point, ContinuousFunctions, Rates of Change: Derivative at a Point, Derivative as a Function, Second- and Higher-Order Derivatives, Instantaneous Rates of Change, Chain Rule, Implicit Differentiation, Tangents and Normal Lines, Linearization, Concavity: The Second Derivative Test for Concavity. **[Relevant topics from: Text 1- sections 2.2, 2.5, 3.1, 3.2, 3.3, 3.4, 3.6, 3.7, 3.9, 4.4]**	9
2	Functions of Several Variables: Domains and Ranges, Level curvesof two variables, Limits for functions of two variables, Continuity for functions of two variables, Partial derivatives of a functions of more than two variables, Partial derivatives and continuity, Second- Order partial derivatives, The mixed derivative theorem, TheChain Rule: Functions of two variables **[Relevant topics from: Text 1- sections 14.1, 14.2, 14.3, 14.4]**	9

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3	The Chain Rule: Functions of three Variables, Directional Derivatives in the Plane, Interpretation of the Directional Derivative, Gradient, Properties of the Directional Derivative, Local Extreme Values for Functions of Two Variables: Relative extrema, First derivative theorem for local extreme values, Critical point, saddle point, Second Derivative Test for Local ExtremeValues, Absolute Maxima and Minima on Closed Bounded Regions. **[Relevant topics from: Text 1- sections 14.4, 14.5, 14.7]**	**9**
4	Constrained Maxima and Minima, The Method of Lagrange Multipliers with one constraint, The Method of Lagrange Multipliers with two constraints, Method of Steepest Descent (only two variables), LPP-Formation, Solution of LPP using graphic method. **[Relevant topics from: Text-1 section 14.8, Text-2 sections 22.1, 22.2]**	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

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End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one fullquestion out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply various concepts in calculus to linearize functions and to analyze concavity.	**K3**
CO2	Calculate the limits for functions of two variables and partial derivatives of multivariable functions.	**K3**
CO3	Interpret directional derivative and solve maxima and minima of multivariable functions.	**K3**
CO4	Solve constrained maxima and minima, LPP and understand the method of Steepest Descent.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate, K6- Create*

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CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Thomas' Calculus	Maurice D. Weir, Joel Hass, Christopher Heil,Przemyslaw Bogacki	Pearson	15th edition, 2023
2	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10th edition, 2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Calculus	Howard Anton, Irl Bivens, Stephens Davis	Wiley	10th edition,2012
2	Optimization: Algorithms and Applications	Rajesh Kumar Arora	CRC Press	1st edition, 2015
3	Multivariable Calculus	Ron Larson, Bruce Edwards	Brooks/Cole, Cengage Learning	10th edition,2014
4	Calculus & Its Applications	Goldstein, Schneider, Lay, Asmar	Pearson	14th edition,2018
5	Bird's Higher Engineering Mathematics	John Bird	Taylor & Francis	9th edition, 2021
6	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39th edition, 2023

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Video Links (NPTEL, SWAYAM…)	
Module No.	**Link ID**
1	https://nptel.ac.in/courses/111106146
2	https://nptel.ac.in/courses/111107108
3	https://nptel.ac.in/courses/111107108
4	https://nptel.ac.in/courses/111107108

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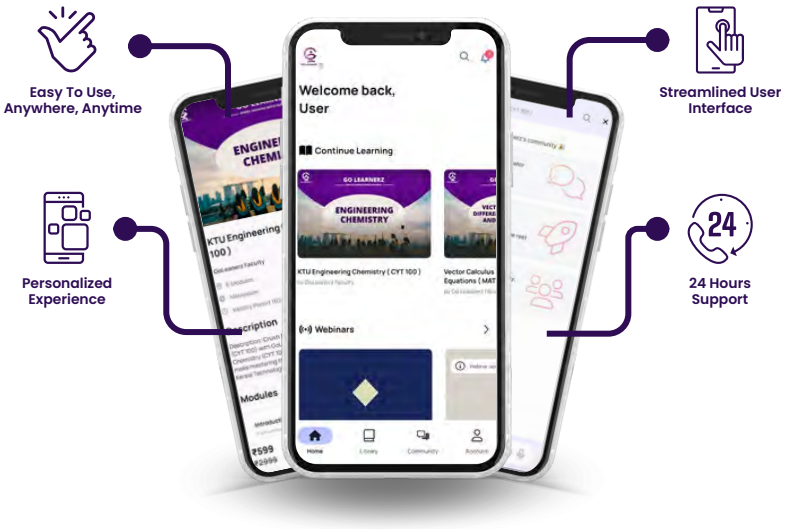

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